

StudyBuddy

CW2 Report

IN3065 User-Centred System Design 2025–2026

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Raw data (interview recordings): [OneDrive folder \(City login required\)](#)

Contents

1	User Research	1
1.1	User Research Methodology	1
1.2	User Research Results	1
1.3	Current User Journey Map	3
1.4	Primary Persona	4
1.4.1	Persona Creation Method	4
2	Design	5
2.1	Wireframes Intro	5
2.2	Flow Diagram	6
2.3	Annotated Wireframes	7
3	Usability Testing Plan	13
	References	14
	Appendices	15
A	Research Materials	15
B	Informed Consent Form Template (Final)	16
C	Participant Information Sheet (Final)	17
D	Interview Guide / Script (Final)	18
E	Full List of Findings	19
F	Interview Summaries	20
G	Evidence of Analysis	21
H	Observation / Walkthrough Notes	22
I	GenAI Usage Log	23

1 User Research

1.1 User Research Methodology

(max 250 words)

Four semi-structured interviews were conducted with potential users of StudyBuddy: undergraduate Y2 Electrical Engineers enrolled on the same semester-2 module set (mechatronics, engineering design, sensors & instrumentation, data analysis, engineer in society).

Recruitment used a five-question Qualtrics prescreening survey (Appendix A) with inclusion criteria: (i) current undergraduate or postgraduate study; (ii) regular study outside contact hours; (iii) use of third-party tools beyond Moodle, course textbooks, and OneDrive; and (iv) availability for a ~20-minute interview. Criterion (iii) deliberately biased recruitment towards students who already supplement Moodle with personal tooling. This is the population StudyBuddy is positioned to support.

Interviews ran 20–30 minutes each, in person, semi-structured, and video-recorded with participants' faces shown. Recordings were transcribed via Teams' auto-transcript function and cleaned to markdown. All four interviews included a walkthrough segment, in three different forms: live screen-share of a study setup (P1, P4); demonstration of physical artefacts, including a wall-mounted colour-coded notes-and-equations board (P2); and a brief verbal scenario walkthrough of a pre-exam study session (P3, less in-depth than the other three, with no live setup or artefact shown).

Ethics followed the module template: informed consent forms collected real names and University email addresses, and a Participant Information Sheet was provided. Participants were anonymised to P1–P4 at the cleaning stage; consent forms are submitted separately.

Analysis followed Braun and Clarke (2006)'s reflexive thematic analysis: familiarisation, line-level open coding (Appendix G), affinity-style clustering into candidate themes, review against data extracts, and final theme definition. Walkthrough observations are reported separately as descriptive records (Appendix H).

1.2 User Research Results

(max 600 words)

Participants

Four undergraduate engineering students at City University, on a shared semester-2 module set.

P1: NotebookLM and Notion user; fixed 20:00–23:00 evening study block; live walkthrough.

P2: heavy hand-writing practice with a wall-mounted formula board; refuses evening study; single desk shared with gaming PC; artefact-only walkthrough.

P3: late-night bedroom study to avoid family activity; minimal tooling; brief verbal walkthrough only (no live setup or artefact).

P4: multi-monitor setup with an iPad subject/exam/topic folder hierarchy; leaves home for serious coursework; live walkthrough.

Key Findings

T1: “Which equation?” The dominant cognitive bottleneck is selecting *which* equation to apply, not arithmetic (P2, line 77; P3, line 57). The most-wanted artefact is a step-by-step worked solution. Where Moodle provides only numerical answers (“560 kg for mass” P1, line 263) or releases

solutions a week late (P4, line 159), participants cascade through nearest-similar tutorial → AI → YouTube → pre-exam peer help. AI use is trust-bounded by an explicit fear of “AI getting the calculation totally wrong — showing you the right result, but not the right way to get there” (P2, line 169).

T2: Prioritisation is personal and rarely tracked. Each participant uses a different primary rule: difficulty (P1), session-count goals (P2), deadline (P3), motivation (P4). Time-of-day preferences are mutually incompatible (20–23h, 14–16h, 22–24h, timetable-gap-only). No participant plans more than ~3 days ahead or tracks progress beyond mental goals. Effort, not output, is the implicit success metric (P2, line 285).

T3: Focus is engineered, not willed. All four participants enter focus through deliberate spatial (bedroom-vs-living-room), temporal (time-of-day), and physical (keyboard relocation, single earbud, wall-plank, multi-monitor) levers. Default willpower is not the strategy.

T4: Group study is exam-shaped, ad-hoc, brittle. All four use group study, only as exams approach, coordinated spontaneously (“@everyone, who wants to join?”), and vulnerable to off-topic chatter and pace mismatch (P1, line 205). It nonetheless performs two real functions: screen-shared problem solving (a substitute worked example) and explain-to-learn (P4, line 247).

T5: Moodle is a delivery channel, not a study artefact. All four treat Moodle as the inescapable on-ramp, but each builds a separate personal external memory beside it — in radically different forms (Notion + NotebookLM; hand-written colour-coded notes plus wall-plank; iPad folder hierarchy; ad-hoc downloads). Friction is unevenly perceived but the underlying pattern is uniform.

Design Goal

Address the **worked-example gap (T1)** and the **personal, short-horizon prioritisation pattern (T2)** that determines when and on what these students actually revise, for second-year undergraduates revising quantitative engineering modules. Specifically: surface step-by-step worked solutions and AI procedure explanation with hallucination guardrails (T1), delivered to fit each student’s own prioritisation rule and ad-hoc, deadline-driven cadence rather than imposing a long-horizon study plan (T2), and *coexisting with* — rather than replacing — the personal external memory each student has already built (T5).

Key Requirements

- Surface step-by-step worked solutions for tutorials and past papers, not just numerical answers (T1; F4, F6).
- Make AI procedure explanation auditable, with explicit addressing of hallucinated-working fear (T1; F11).
- Operate inside an ad-hoc, deadline-driven cadence; do not impose long-horizon planning (T2; F21–F22).
- Adapt to the student’s own prioritisation rule (difficulty, deadline, session-count, motivation) rather than imposing a single ranking (T2; F18–F20).
- Coexist with, not replace, students’ existing external memory artefacts (T5; F46).

1.3 Current User Journey Map

1.4 Primary Persona

1.4.1 Persona Creation Method

(max 200 words)

2 Design

2.1 Wireframes Intro

2.2 Flow Diagram

2.3 Annotated Wireframes

Wireframe 1: [screen name]

Wireframe 2: [screen name]

Wireframe 3: [screen name]

Wireframe 4: [screen name]

Wireframe 5: [screen name]

Wireframe 6: [screen name]

3 Usability Testing Plan

(max 450 words)

Evaluation Aim

Participants

Environment & Equipment

Data Captured

Task Scenarios

Procedure

References

Braun, Virginia and Victoria Clarke (2006). "Using thematic analysis in psychology". In: *Qualitative Research in Psychology* 3.2, pp. 77–101. DOI: 10.1191/1478088706qp063oa.

A Research Materials

B Informed Consent Form Template (Final)

C Participant Information Sheet (Final)

D Interview Guide / Script (Final)

E Full List of Findings

F Interview Summaries

G Evidence of Analysis

H Observation / Walkthrough Notes

I GenAI Usage Log

The following table documents all use of Generative AI tools during the preparation of CW2 materials, including prompts/inputs, outputs and uses.

Date	Tool		Prompt/Input	Output & Use
2026-04-23	Claude 4.7	Opus (Claude Code CLI)	start with a skeleton for pt2, and sections to logically work through.	Created part-2/pt2.tex as an empty skeleton mirroring the CW2 brief structure (cover page, User Research / Design / Usability Testing sections, References, Appendices and this GenAI log) and an empty references.bib. All section content left as TODO comments for the student to author.
2026-04-24	Claude 4.7	Opus (Claude Code CLI)	Write this to a md file for context for me.	Wrote part-2/ROADMAP.md as a personal working document (not a submission deliverable) capturing locked design direction, the planned wireframe journey, an analysis pipeline outline, a phase-by-phase checklist and a marking-weight table. No analytic content (findings, persona, wireframes) was generated.

Date	Tool	Prompt/Input	Output & Use
2026-04-25	Claude Opus 4.7 (Claude Code CLI)	In Coursework I need to do some analysis of the interviews. (with follow-up corrections redirecting to the CW2 brief 1:1 and to thematic analysis as the single method.)	Drafted part-2/analysis.md as a personal working analysis document (not submitted) applying Braun & Clarke (2006) reflexive thematic analysis across the four cleaned transcripts: method block, participant table, per-interview summaries, line-level open codebooks, a code-cluster theme map, five named themes (T1–T5) with pull quotes, and a granular findings list F1–F49. All quotations cited by interview file and line; no content added to pt2.tex beyond this log entry.
2026-04-26	Claude Opus 4.7 (Claude Code CLI)	correct the onedrive folder url properly please. + can you take a look at my pt2? + clean all these up.	Cover-page OneDrive placeholder replaced with a clean \href link. Cleanup pass on pt2.tex (no analytic content added): fixed a stray back-tick in the URL; escaped a bare & in §1.1; repointed two undefined \ref targets; added a Research Materials appendix as the resolution target for the Qualtrics-screener cite; deleted four duplicate empty subsection stubs in §1.2; added a Braun & Clarke 2006 entry to references.bib and converted the inline citation to \textcite{braun2006} so the bibliography is populated.

Date	Tool	Prompt/Input	Output & Use
2026-04-26	Claude Opus 4.7 (Claude Code CLI)	tbf participant 3 did do a mini walkthrough but not as in depth as the others. (student-supplied excerpt from Interview3.md.)	Corrected the walkthrough count in §1.1 from <i>three of four</i> to <i>all four</i> , and rewrote P3's blurb in §1.2: P3 gave a brief verbal scenario walkthrough only (no live setup or artefact). Mirror updates applied to analysis.md (method block, participant table, §7 intro, and a new §7.4 with the verbal-walkthrough notes). The thematic analysis itself was carried out by the student.